Fields and Waves I

Lecture 10

Gauss's Law

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Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Overview

- Gauss's Law



Review

Gauss's Law

- Points that contain charge have a non-zero divergence, whereas the points with no charge have a divergence of zero.

This is the differential form of Gauss's Law.

$$\nabla \cdot \vec{D} = \rho$$

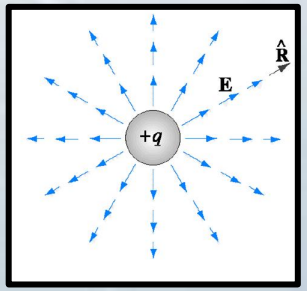
$\vec{D} = \varepsilon \vec{E}$

ε: permittivity

- The electric field flux through a closed surface is determined by the amount of charge inside. This can be rewritten as:

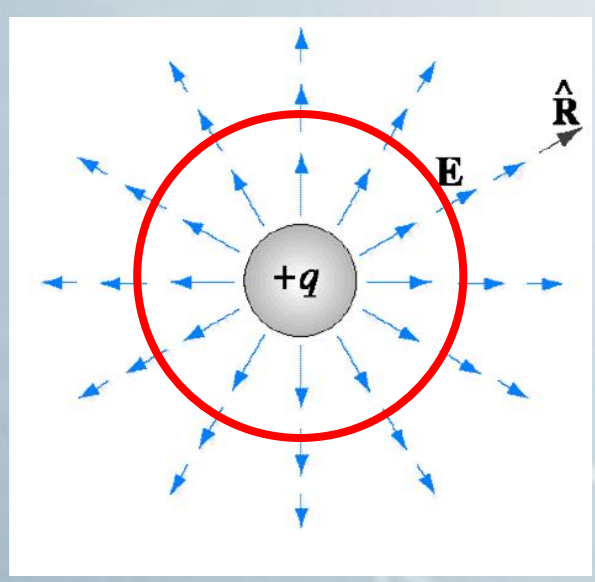
This is the integral form of Gauss's Law.

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_{enc}}{\epsilon}$$



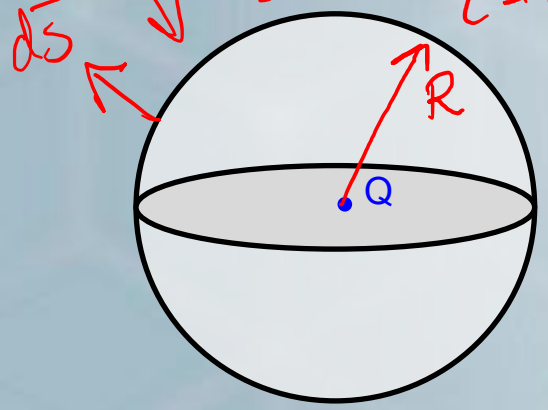
Electrostatics

- Show that the electric field of a point charge satisfies Gauss' Law by integrating $\oint \vec{E} \cdot d\vec{S}$ over the surface of a sphere of radius a.



$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2}\hat{R}$$

Electrostatics



$\varepsilon = \varepsilon_0$ ← constant $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2}\hat{\boldsymbol{a}}_r$$

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_{enc}}{\epsilon}$$

$\iiint \rho dV / \varepsilon$

RHS: $Q_{enc}/\varepsilon_0 = Q/\varepsilon_0$ (charge integral)

LHS: $\oint \vec{E} \cdot d\vec{S}$ (flux integral)

$$= \int_0^{2\pi}\int_0^{\pi} \left(\frac{Q}{4\pi\varepsilon_0 r^2}\hat{r}\right) \cdot \left(r^2 \sin\theta \, d\theta \, d\phi \, \hat{r}\right) = \frac{Q}{4\pi\varepsilon_0}\int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta \, d\theta$$

$$= \frac{Q}{4\pi\varepsilon_0}(2\pi)\cdot(2) = Q/\varepsilon_0$$

Electrostatics: Coulomb's Law

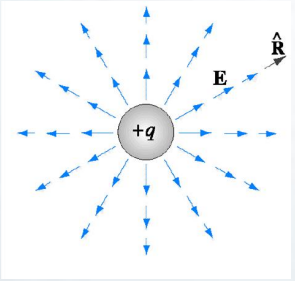


Electrostatics

Coulomb's Law

Coulomb's Law gives the force between two, stationary, charged particles

$\vec{E}$,of Q_1 is $\frac{Q_1}{4\pi\varepsilon_0 R^2}\hat{\hat{R}}$



Unit vector pointing away from Q_1

Then, $\vec{F_{12}} = Q_2 \cdot \vec{E}$

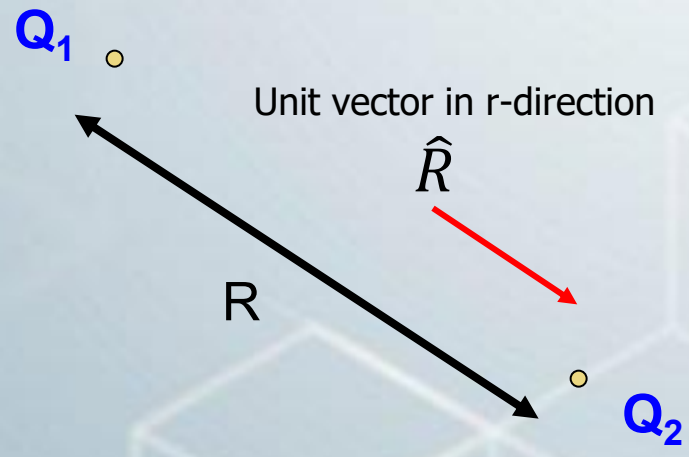
Electrostatics

Coulomb's Law

$\vec{F}$ (force), between point charges

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R^2}\hat{R}$$

(force on Charge 2 by Charge 1)



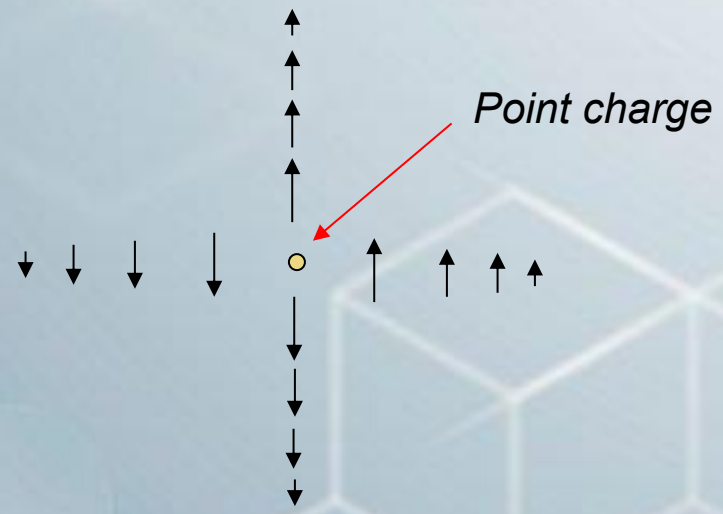
Electrostatics

$\vec{E}$, is a **VECTOR** Field

How do we represent it?

The electric field points in the direction that a +q test charge would move

Represent using Arrows : Direction and **Length**



Proportional to **magnitude** or strength of E-Field

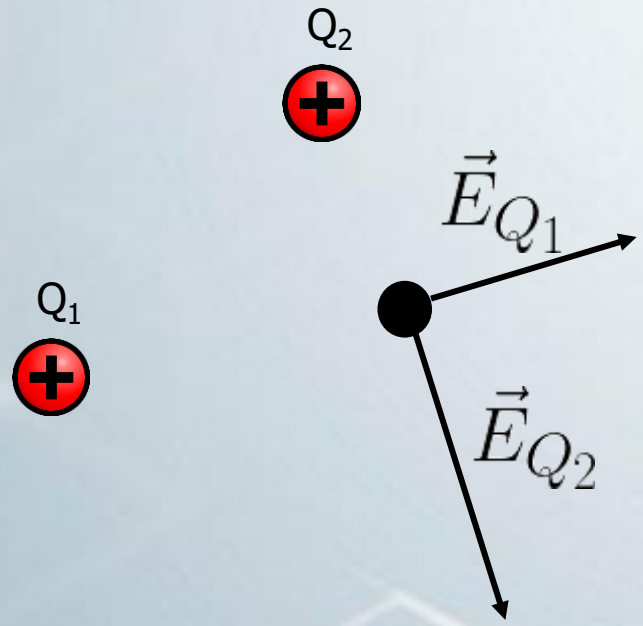
Electrostatics

- To compute E-fields from multiple charges, apply **superposition of fields**.



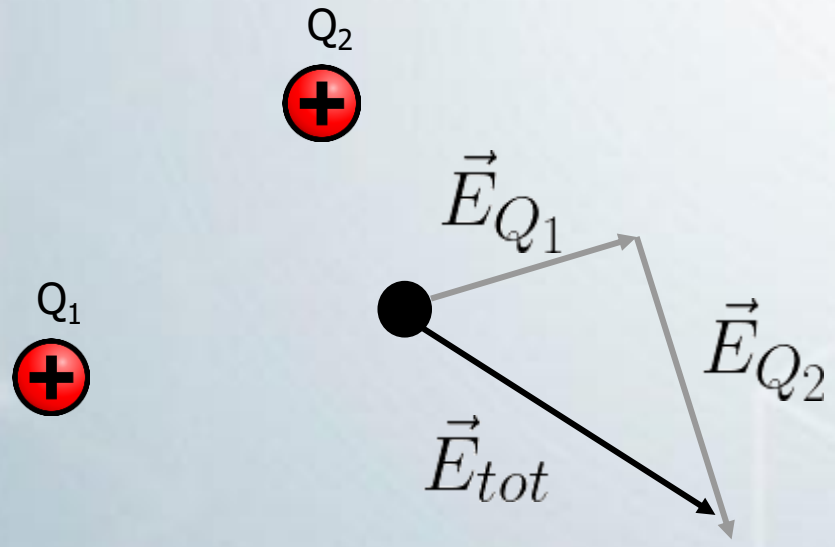
Electrostatics

- To compute E-fields from multiple charges, apply **superposition of fields**.



Electrostatics

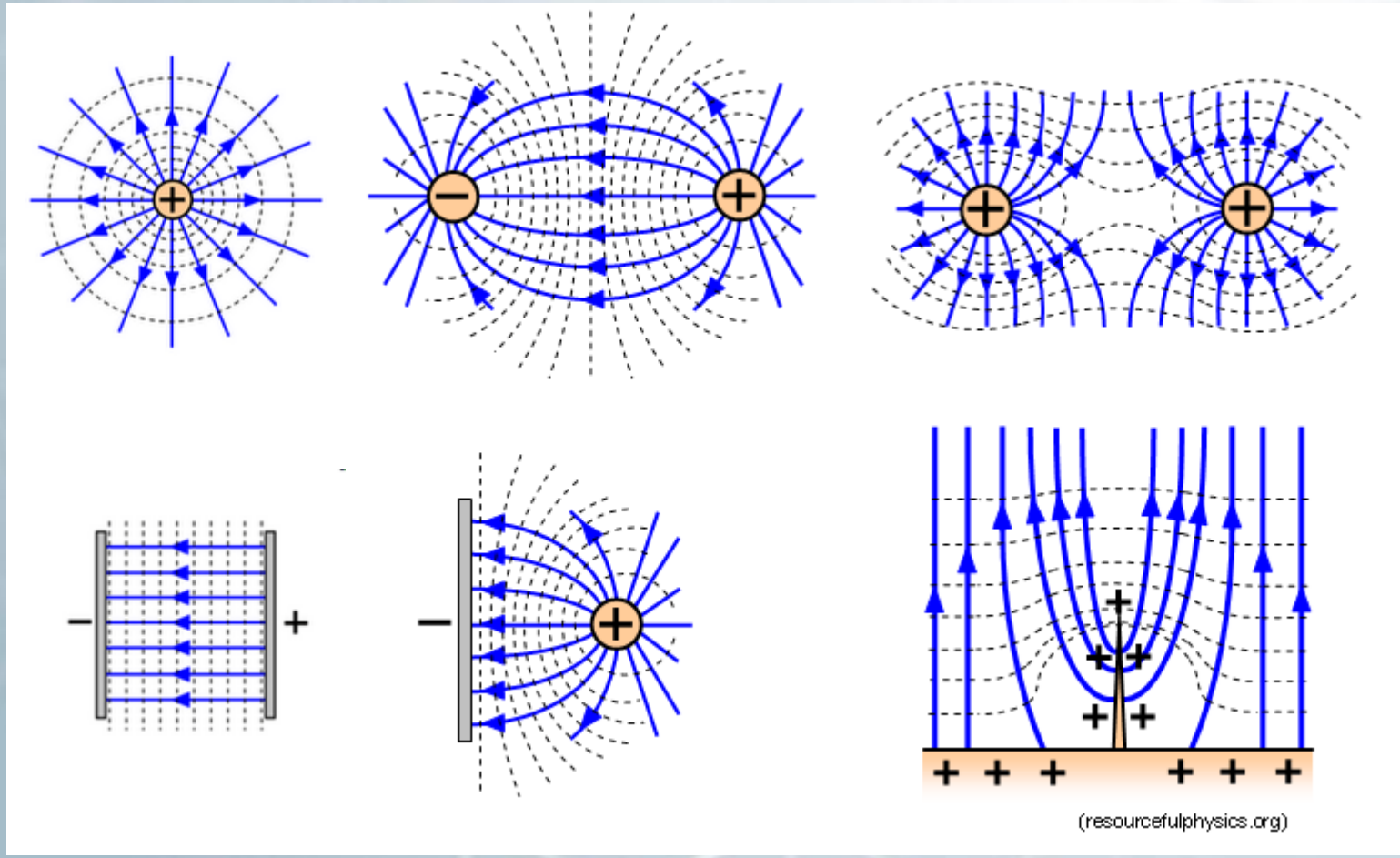
- To compute E-fields from multiple charges, apply **superposition of fields**.



Do Lecture 10 Exercise 1 in groups of up to 4.

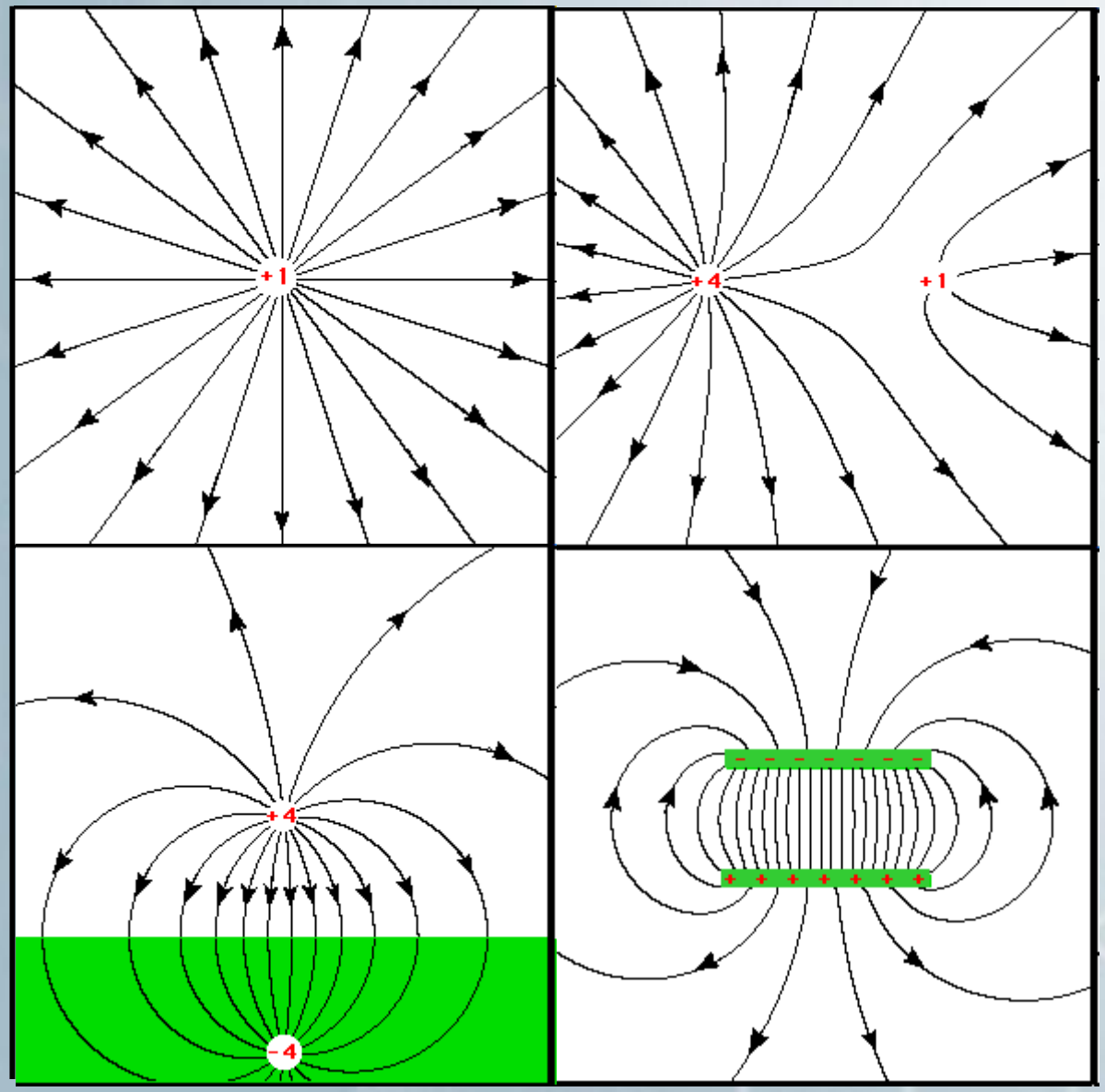
Electrostatics

More examples of electric fields and charge distributions



http://shinliang.blogspot.com/2009/04/21-coulombs-law.html

Electrostatics



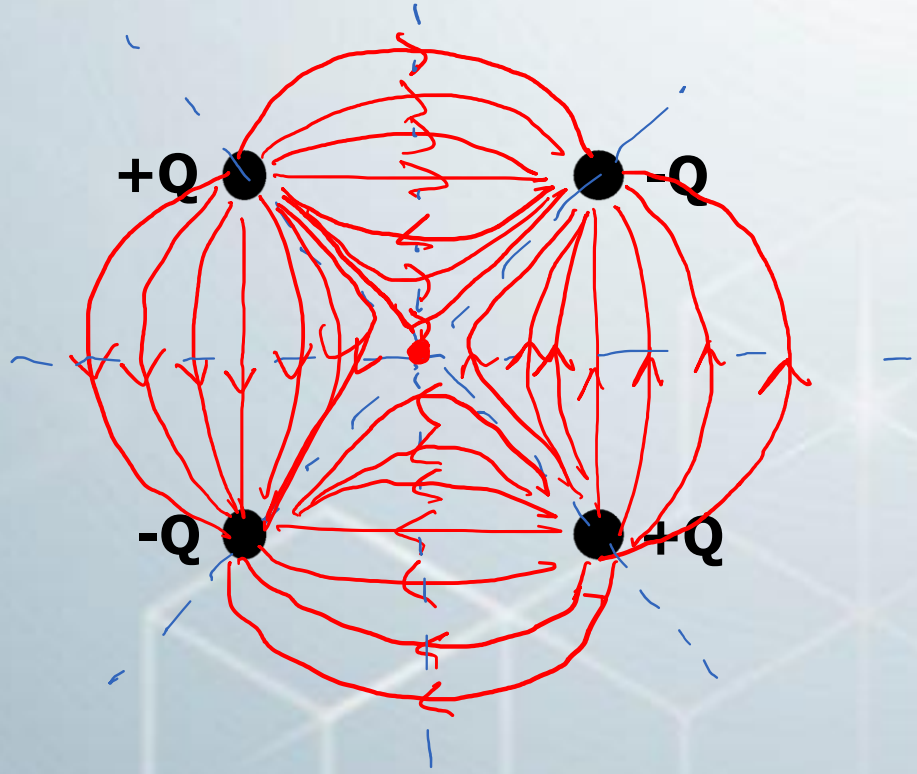
http://people.seas.harvard.edu/~jones/cscie129/nu_lectures/lecture6/field_vis/e_vis.html

- Note, in the upper right figure, that four times as many field lines leave the +4 positive charge as leave the +1 charge.

- All of the field lines end at infinity, as they do with a single positive charge (E field is defined in ***all*** space).

Electrostatics

- Sketch the electric field lines for the electric quadrupole shown. Sketch the planes for which you expect the field to be symmetric. *After completing your sketch,* verify your result with the applet at https://davidawehr.com/projects/electric_field.html



Electrostatics: Gauss's Law



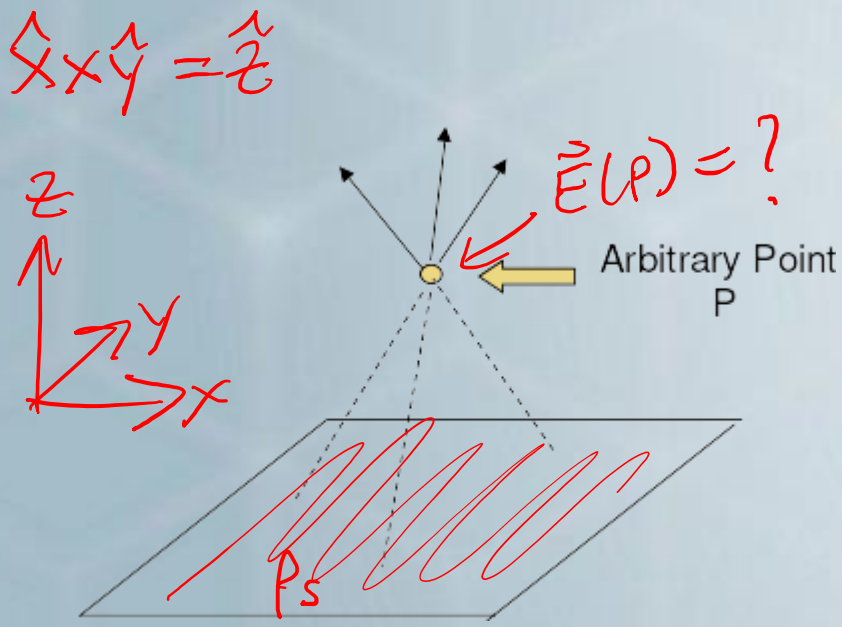
Gauss's Law

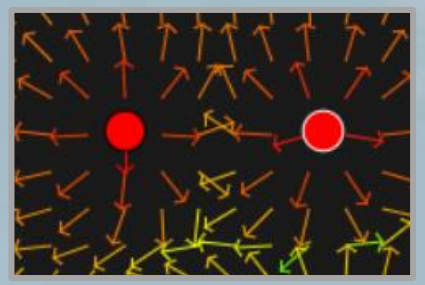
$$\vec{D} = \epsilon \vec{E}$$

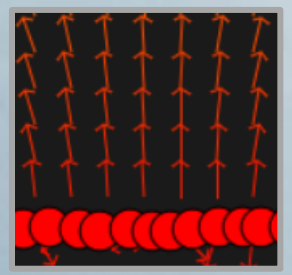
- Previously we said: $\oint \vec{D} \cdot ds = \int \rho \cdot dv = Q_{enc}$

- Can we understand this in terms of electric field lines?

Gauss's Law





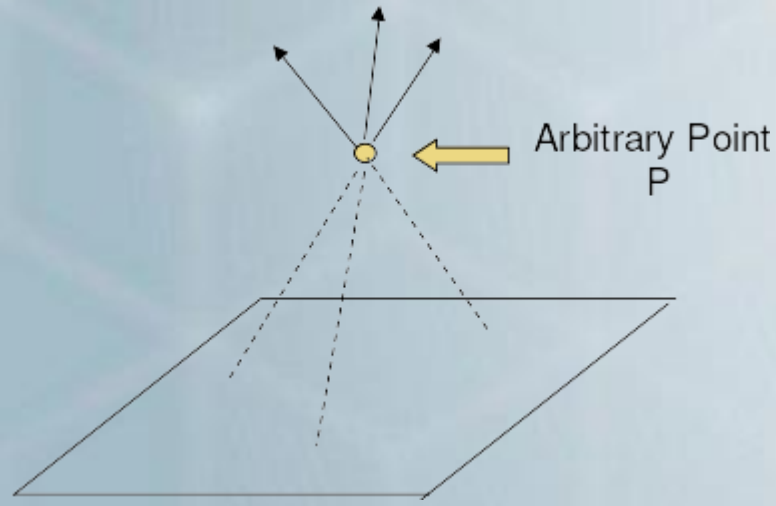


- What would the **electric field** be at an arbitrary point suspended over an infinite surface of uniform charge per area (ρ_s in C/area)?
- First question: what direction will the field point in?
- The +z direction since all non-z components will cancel.

$$\vec{E} = E_z(z)\hat{z}$$

$$\vec{E}(x,y,z) = \langle E_x(x,y,z), E_y(x,y,z), E_z(x,y,z) \rangle$$

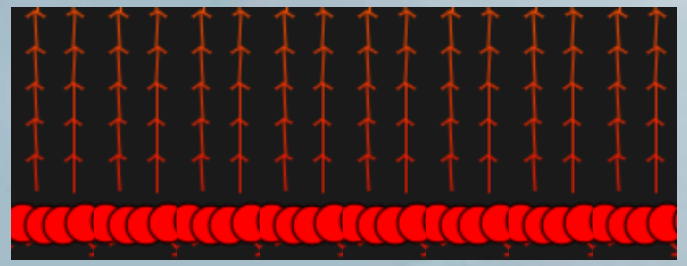
Gauss's Law



- Second question: should the field be uniform everywhere above the surface at the **same z**?

Yes! **(more on this later)**

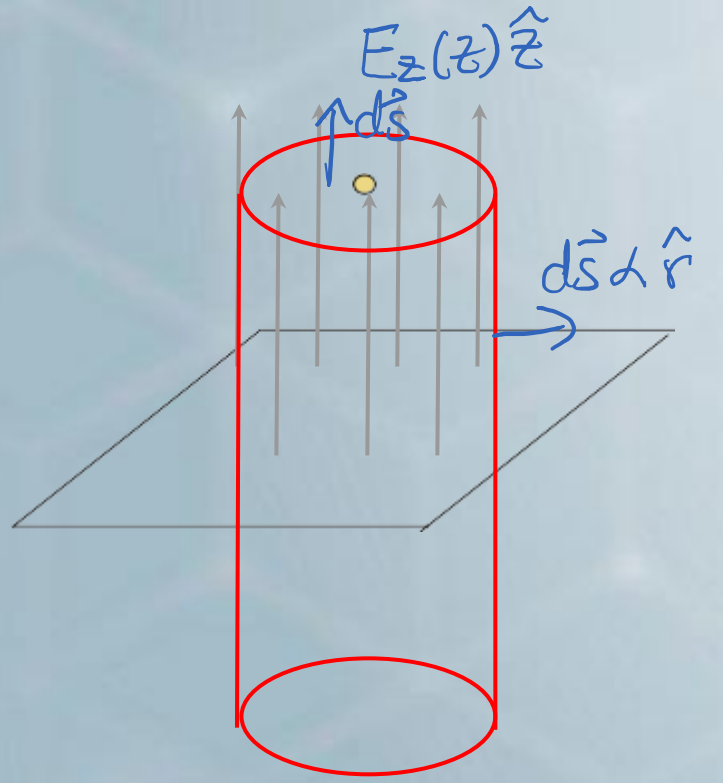
$$\vec{E} = E_z(z)\hat{z}$$



Gauss's Law

$$\hat{z}\cdot\hat{r}=0$$

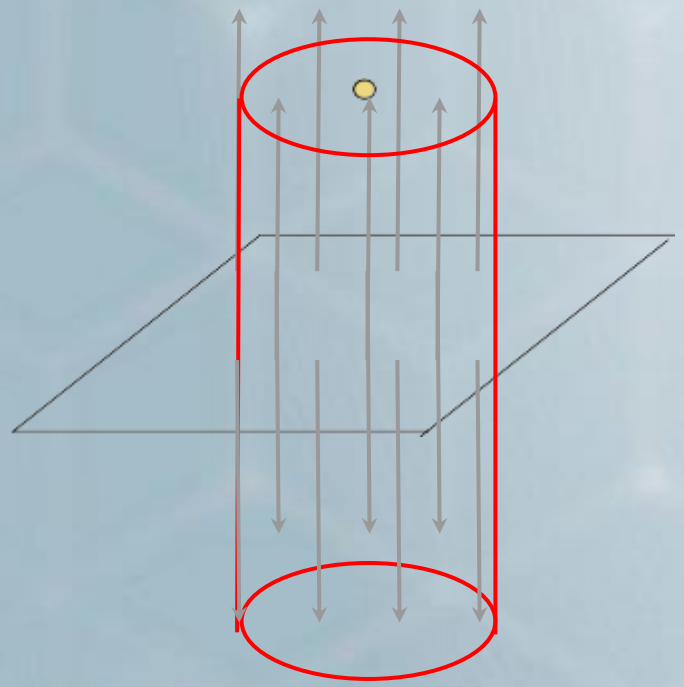
$$\iint \vec{E}\cdot d\vec{s} = \iint_{top} E_z(z)\hat{z}\cdot ds\hat{z} + \iint_{side} E_z(z)\hat{z}\cdot ds\hat{r}$$



- Third question: If we make a Gaussian surface and measure the flux through it, which surfaces have a non-zero net flux?

The ends!

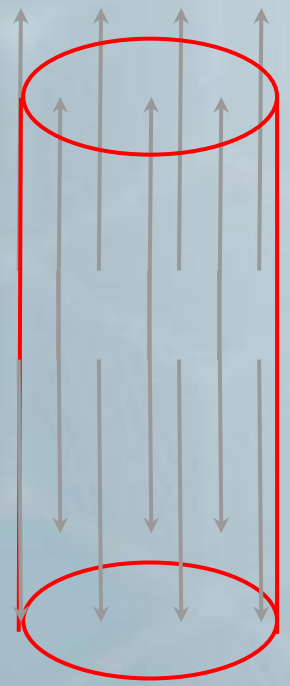
Gauss's Law



- Fourth question: Are there any lines we forgot to draw?

Yes, the equal and opposite ones on the underside!

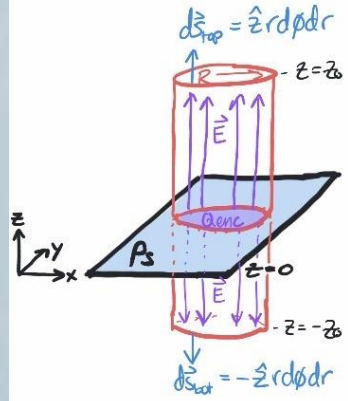
Gauss's Law



- Fifth question: What is the electric field as calculated by Gauss's law for this surface?

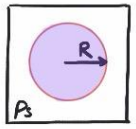
Gauss's Law

Gauss's Law #1: Sheet of Charge



$$\oint \vec{D}\cdot d\vec{s} = Q_{enc}$$

- What is Q_{enc}?



$$Q_{enc} = \int_0^{2\pi}\int_0^{R} \rho_s\, r\,dr\,d\phi = 2\pi\rho_s \int_0^{R} r\,dr = \pi R^2 \rho_s$$

- What is $\oint \vec{D}\cdot d\vec{s}$?

$$\text{Flux} = \oiint \vec{D}\cdot d\vec{s} = \iint_{top} \varepsilon_0 \vec{E}\cdot d\vec{s}_{top} + \iint_{bottom} \varepsilon_0 \vec{E}\cdot d\vec{s}_{bot} + \iint_{side} \varepsilon_0 \vec{E}\, d\vec{s}_{side}$$

($\vec{E}$ is perpendicular to $d\vec{s}_{side}$, so the side integral is 0)

$$= \varepsilon_0 \int_0^{2\pi}\int_0^{R} (E_z(z_0)\hat{z})\cdot(r\,dr\,d\phi\,\hat{z}) + \varepsilon_0 \int_0^{2\pi}\int_0^{R} (-E_z(-z_0)\hat{z})\cdot(-r\,dr\,d\phi\,\hat{z})$$

$$= 2\varepsilon_0 \int_0^{2\pi}\int_0^{R} E_z(z_0)\, r\,dr\,d\phi = 2\varepsilon_0 \cdot 2\pi \cdot E_z(z_0)\frac{R^2}{2} = \pi\varepsilon_0 R^2 E_z(z_0)$$

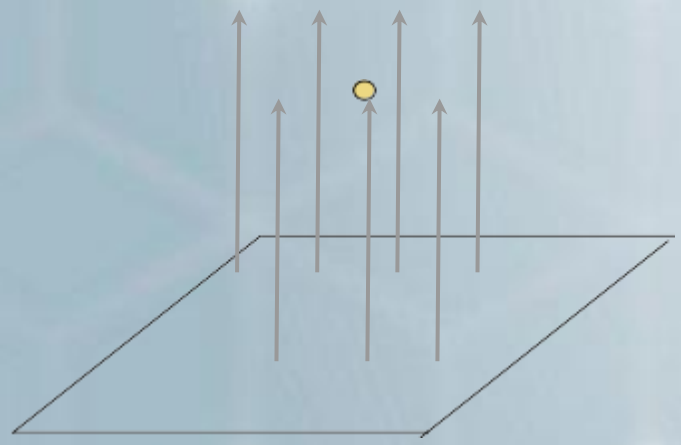
due to symmetry $|E_z(z_0)| = |E_z(-z_0)|$

- Equate total flux with enclosed charge

$$2\pi\varepsilon_0 R^2 E_z(z_0) = \pi R^2 \rho_s \Rightarrow E_z(z_0) = \frac{\rho_s}{2\varepsilon_0}$$

$$\vec{E}(z) = \begin{cases} \rho_s/2\varepsilon_0\,\hat{z} & \text{for } z>0 \\ -\rho_s/2\varepsilon_0\,\hat{z} & \text{for } z<0 \end{cases}$$

Gauss's Law

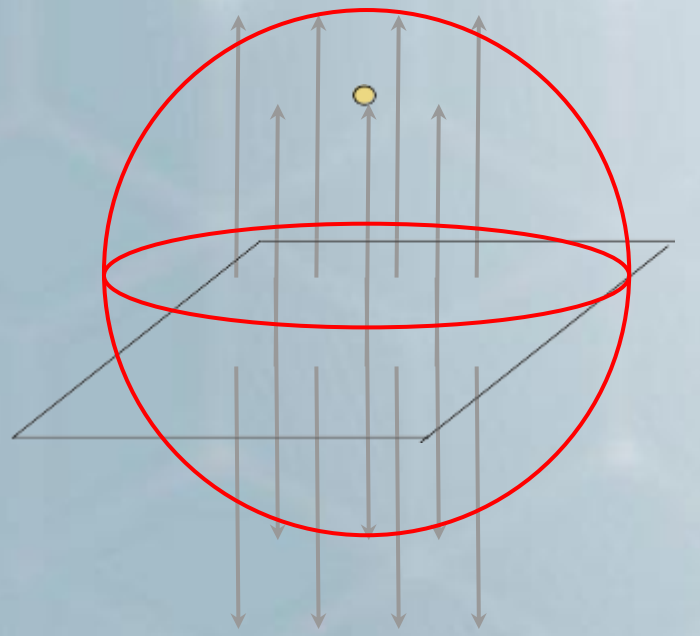


$$\vec{E} = \frac{\rho}{2\varepsilon_0}\hat{z}$$

- This is an interesting answer because we know that field strength generally falls of with distance, and this expression has no dependence on distance at all. Does this make sense?

Yes… if the plane is truly infinite, it should look exactly the same no matter how high above it you are because there is an infinitely large distribution of charge.

Gauss's Law



- Sixth question: Will this be the same for any Gaussian surface we choose?

Yes!

Gauss's Law

Steps to solve problems using Gauss's Law

1. Recognize the coordinate system.
2. Using symmetry, determine which components of the field exist.
3. Identify a Gaussian surface for which the sides are either parallel to or perpendicular to the field components. This surface is arbitrary in size.
4. Determine the total charge within that surface. The charges can be distributed on lines, surfaces or in volumes.

Gauss's Law

5. To evaluate the electric flux passing through the Gaussian surface.

   - If the field is parallel to the surface $\oint \mathbf{D} \cdot \mathrm{d}\mathbf{S} = 0$

   - If the field is constant and perpendicular to the surface (surface area A),

$$\oint \mathbf{D} \cdot \mathrm{d}\mathbf{S} = D_i \oint \mathrm{d}A_i = D_i A_i$$

     where the subscript refers to the direction of the surface normal.

   - Note that a high level of symmetry is necessary to make these simplifications.

Gauss's Law

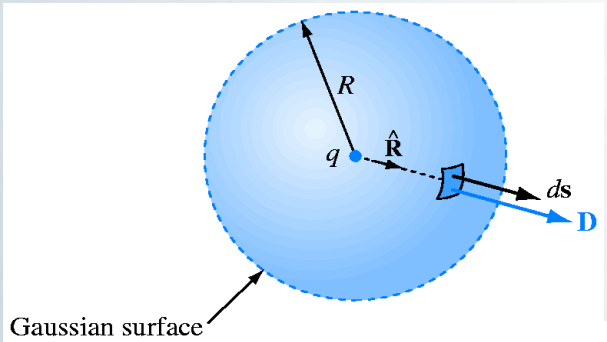
6. Now equate the two sides of Gauss' Law to find E:

$$D_i A_i = \oint \mathbf{D} \cdot \mathrm{d}\mathbf{S} = \int \rho \mathrm{d}V = Q_{enc}$$

Slido!

Gauss's Law

- Examples of typical Gaussian surfaces



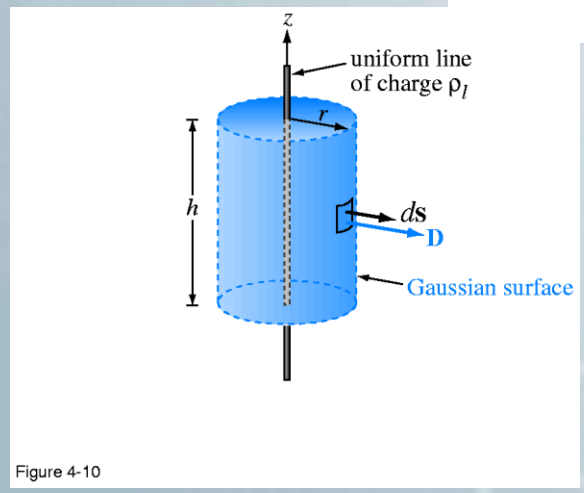
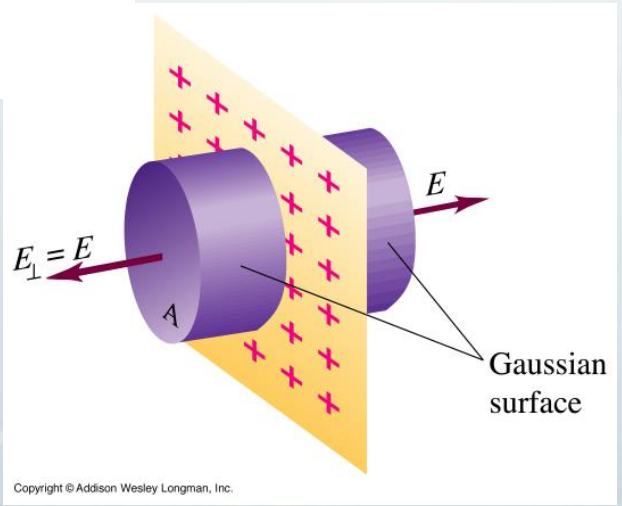


Figure 4-10



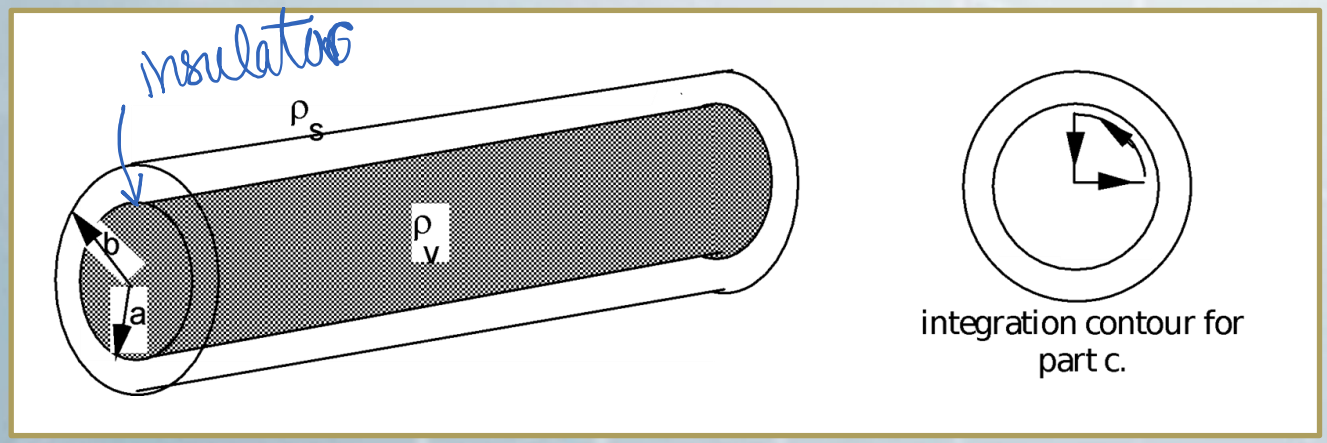
Ulaby

Gauss's Law

Full Gauss' Law Solution

A charge distribution with *cylindrical* symmetry is shown. The inner cylinder has a uniform **volumetric charge density** $\rho_v\ [C/m^3]$

The outer shell has a **surface charge density** $\rho_s\ [C/m^2]$ such that the total charge on the outer shell is the negative of the total charge in the inner cylinder. Ignore end effects.

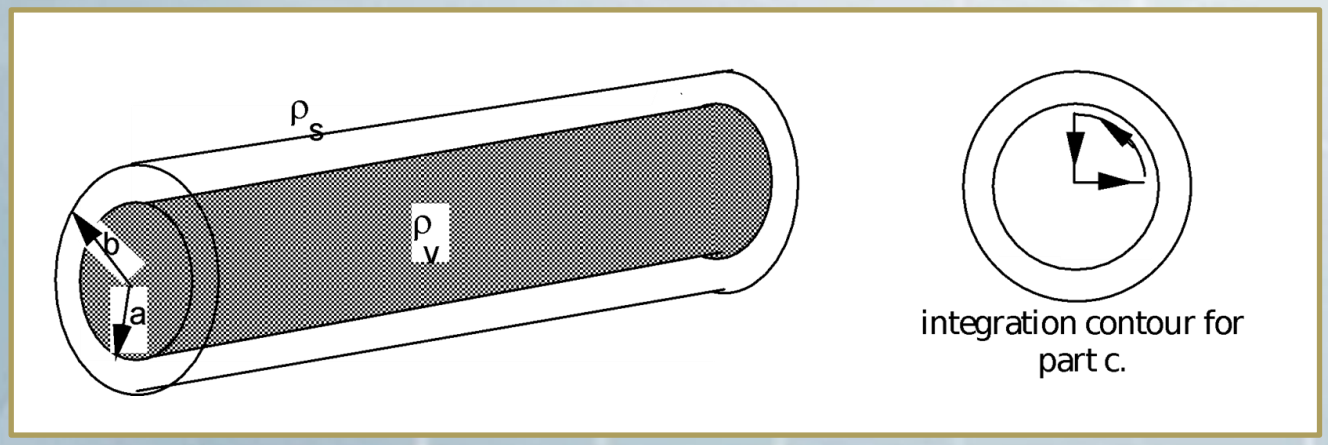


Gauss's Law

1. Find the electric field for all *r*.

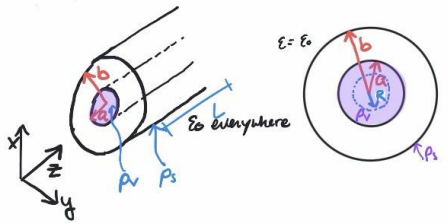
2. Check your answer by evaluating the divergence and curl of the electric field.

3. What is the closed line integral of the electric field around the contour shown?



Gauss's Law

Gauss's Law #2: Shielded Cable



Symmetry: $\vec{E}(\vec{r}) = E_r(r)\hat{r}$

1) for $0 \le r < a$: $\oint \vec{D} \cdot d\vec{s} = Q_{enc}$

$$Q_{enc} = \int_0^L \int_0^{2\pi} \int_0^R \rho_v \, r \, dr \, d\phi \, dz = L \cdot 2\pi \frac{R^2}{2} \rho_v = \pi R^2 L \rho_v$$

$$\text{Flux} = \int_0^L \int_0^{2\pi} (\varepsilon_0 E_r(R)\hat{r}) \cdot (R \, d\phi \, dz \, \hat{r}) = \varepsilon_0 \cdot L \cdot 2\pi E_r(R) \cdot R$$

$$\pi R^2 L \rho_v = \varepsilon_0 L \, 2\pi E_r(R) R \rightarrow E_r(R) = \frac{\rho_v}{2\varepsilon_0} R \rightarrow \vec{E}(r) = \frac{\rho_v}{2\varepsilon_0} r \hat{r} \text{ for } 0 < r < a$$

2) for $a \le r \le b$: $\oint \vec{D} \cdot d\vec{s} = Q_{enc} = \int \rho_v dV$

$$Q_{enc} = \int_0^L \int_0^{2\pi} \int_0^a \rho_v \, r \, dr \, d\phi \, dz = L \cdot 2\pi \cdot \rho_v \cdot \frac{a^2}{2}$$

$$\text{Flux} = \varepsilon_0 \cdot L \cdot 2\pi E_r(R) \cdot R$$

$$L \cdot 2\pi \cdot \rho_v \cdot \frac{a^2}{2} = \varepsilon_0 \cdot L \cdot 2\pi E_r(R) \cdot R$$

$$\vec{E}(r) = \frac{a^2 \rho_v}{2\varepsilon_0} \frac{1}{r} \hat{r} \text{ for } a < r < b$$

3) for $r > b$ $\oint \vec{D} \cdot d\vec{s} = Q_{enc} = \int \rho_v dV$

$Q_{enc} = Q_{inner} + Q_{outer} = 0$

$\oint \vec{E} \, d\vec{s} = 0$, so

$\vec{E} = 0$ for $r > b$

4) full solution

$$\vec{E}(r) = \begin{cases} \frac{\rho_v}{2\varepsilon_0} r \hat{r} & \text{for } 0 < r < a \\ \frac{a^2 \rho_v}{2\varepsilon_0} \frac{1}{r} \hat{r} & \text{for } a \le r < b \\ 0 & \text{for } r > b \end{cases}$$

Gauss's Law

- Divergence and curl expressions for cylindrical coordinates:

$$\text{Divergence } \vec{\nabla}\cdot\vec{F} = \frac{1}{r}\frac{\partial(r\cdot F_r)}{\partial r} + \frac{1}{r}\frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}$$

$$\text{Curl } \vec{\nabla}\times\vec{F} = \left(\frac{1}{r}\frac{\partial F_z}{\partial \phi} - \frac{\partial F_\phi}{\partial z}\right)\hat{r} + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r}\right)\hat{\phi} + \frac{1}{r}\left(\frac{\partial(r\cdot F_\phi)}{\partial r} - \frac{\partial F_r}{\partial \phi}\right)\hat{z}$$

- Divergence of $\vec{D}$:

$$0 < r < a : \vec{\nabla}\cdot\left(\epsilon_0\vec{E}\right) = \frac{\frac{1}{r}\partial\left(\frac{\rho_v r^2}{2}\right)}{\partial r} = \rho_v$$

$$a \le r \le b : \vec{\nabla}\cdot\left(\epsilon_0\vec{E}\right) = \frac{\frac{1}{r}\partial\left(\frac{\rho_v a^2}{2}\right)}{\partial r} = 0$$

$$r > b : \vec{\nabla}\cdot\left(\epsilon_0\vec{E}\right) = \frac{\frac{1}{r}\partial(0)}{\partial r} = 0$$

- Curl of $\vec{D}$:for the curl to be non-zero, the $\hat{r}$ component of our electric field would need to contain a dependence on the coordinate ϕ or z. Since it does not, the curl is zero.

Wrap-Up

- HW4 due Wednesday